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Abstract

This paper examines the relationship between government expenditure and GDP growth, using panel data from 10 OECD countries over 20 years. According to neoclassical growth models, technological progress determines long-term growth rates. In his theory of growth, Barro (1990) considers the impact of tax-funded government services on economic growth. Although many empirical studies have been conducted to measure the effects of government spending on economic growth, the relationship remains highly ambiguous. Therefore, this study analyzes whether government expenditures have any effect on GDP growth per capita from the perspective of 10 developed countries. According to our findings, there appears to be a negative impact on growth resulting from public social spending, which is highly significant. Similarly, the results show that government spending on education follows the same pattern.

1. Introduction

Understanding growth has been one of the essential topics amongst economists, and many different theories and ideas about the determinants of growth have been put forward and discussed for decades in the economic literature. The popular neoclassical growth models indicate that the level of technological progress defines long-term growth rates. Unlike the usual neoclassical growth models, Barro (1990) explores the dynamics of growth by addressing the role of tax-financed government services in the equation. According to Barro's model, the possible effect of government spending depends on the type of expenditures. For instance, If the government were to increase spending on productive expenditures, it could positively impact growth. However, if the increase in spending occurs on the unproductive side, it would have a negative effect on growth. Nevertheless, the definition of productive and non-productive government expenditure is not definitive and may vary between researchers' empirical models. In general, productive government spending includes investments in infrastructure, education, and health, while unproductive government spending might include consumption spending such as transfer payments and generalized subsidies. However, government spending on defence and education might be exceptions to this. Because the spending on education and defence are similar to the public investment, hence can make a positive impact on growth by increasing the private sector productivity. But these types of spending depend on the characteristics of the countries, and also their effects might not be seen easily in the short term.

Barro's (1991) research delved into data from 98 countries spanning 1960-1985. His findings revealed a noteworthy inverse correlation between the share of government consumption on GDP and GDP per capita growth. Additionally, he found no substantial relationship between the share of public investment in GDP and GDP per capita growth. It's worth emphasising that although government spending can contribute to economic growth, there exists a threshold beyond which it becomes counterproductive. This is a pivotal aspect of Barro's growth model. Beyond this threshold, increased spending has a diminishing impact and can even impede growth. This relationship is shown in Figure 1. Three main reasons can explain this relationship between government spending and growth. One of the reasons for the

decreasing benefits of government

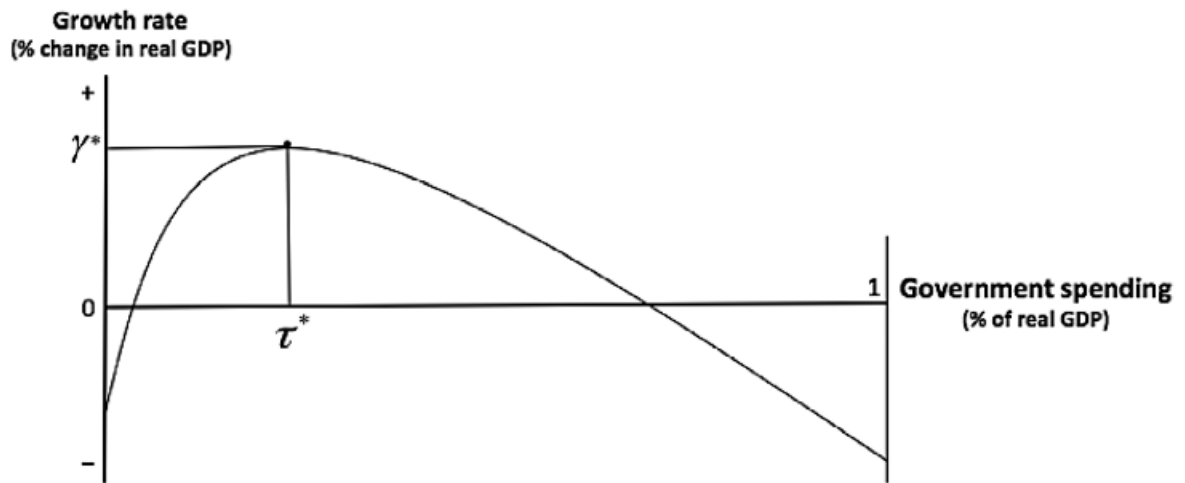


Figure - 1: Government Spending and Growth Rate (Barro, 1990)

spending stems from the diminishing marginal returns to capital, which means that after some threshold, the costs of spending outweigh its benefits because the government needs to finance its expenditure by raising taxes or increasing its debt levels. Secondly, rising government spending can decrease the incentives for private investment since the increased government involvement in the economy can have crowding-out effects. Finally, the connection between growth and spending relies on the type of government expenditures. The majority of the resources allocated to productive causes can promote growth, whilst the allocation mostly devoted to non-productive spending can decrease growth. In that sense, only increasing productive government spending is not a definitive solution to promote growth because the spending should be related to the productive capacity of an economy. Additionally, Chu et al. (2018) showed that a shift in the composition of government spending from non-productive to productive is associated with higher growth in high and low to middle-income countries. The researchers' findings also indicated the potential for

increased government spending to impact the economy through crowding-out effects.

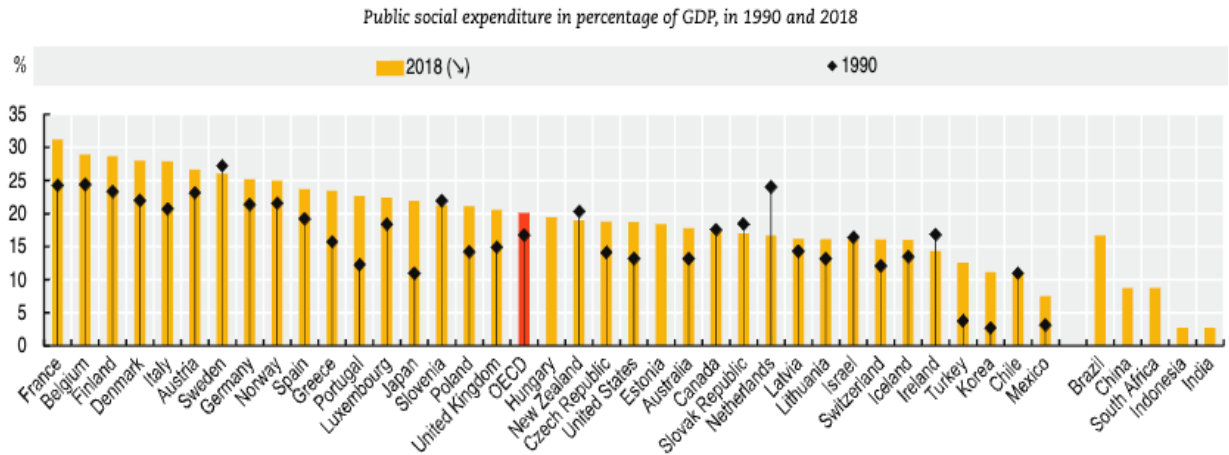


Figure- 2: Source: OECD (2019), Social Expenditure Database (SOCX)

However, over the last decades, there has been clearly an uptrend in the share of public social spending as a percentage of GDP in most of the OECD countries (Figure 2). Interestingly, some rich countries, such as the Netherlands and Ireland, decreased their shares of public social spending as a percentage of GDP by being contrarian to the majority. Given this situation, trying to understand the role of public social spending on GDP growth became worthwhile, especially in the case of most developed countries. According to Connolly and Li (2016), government consumption spending and public investment do not have any significant effects on growth, while public social spending has a significant negative effect. In other words, this might mean that some rich countries can stimulate extra growth by decreasing their utility-type expenditures like public social spending. Therefore, this paper will analyze the impact of public social spending on GDP per capita Growth by utilizing the panel data of the ten most developed countries between the 2002 and 2021 periods.

As a result of our study, we found that public social spending in the developed OECD countries plays a significantly negative role in the GDP per capita growth. Our findings confirm the growth model that was developed by Barro (1990), which suggests that non-productive expenditures such as consumption spending on subsidies and transfer payments have growth-reducing effects.

2. Literature Review

Many different theoretical growth models somewhat influence the literature on economic growth models, but the influence of Barro's (1990) model can be seen through the many empirical models that deal with the relationship between fiscal policy and growth. In fact, some researchers have done studies to understand the causal relationship between fiscal tools and GDP growth. Wu et al. (2010) applied a Granger causality test to 182 countries and found that government expenditure has a role in economic growth. They also reached the conclusion that the effects of government spending in poor countries are usually negative or irrelevant because of poor institution quality and corruption. Guerrero and Parker (2012) discovered that causality flows from GDP growth to government spending and that an increase in GDP growth speeds up the growth of government spending. Conversely, Ahuja and Pandit (2020) demonstrate that causality only flows from government expenditure to GDP growth. Nonetheless, they observe a favorable and significant effect of government spending on the growth of developing countries. Also, as noted by Schaltegger and Torgler (2006), there is evidence of Granger causality from government size to GDP growth.

Kneller et al. (1999) investigated whether there is any empirical evidence supporting the predictions of endogenous growth models claiming that taxation and government spending may have an impact on the steady-state growth rate. The scholars used panel data from 22 OECD countries covering the period from 1970 to 1995, obtained from the Government Financial Statistics Yearbook and the World Bank Tables. The dataset included variables outlining government revenue and expenditure, with a further categorization of expenditure as productive or non-productive. The authors utilized five different regression methods in their model, but the most efficient proved to be the fixed effects model, which included country and time dummies. Their findings exhibited robust outcomes indicating that productive government spending positively influences growth, while distortionary taxation has adverse effects. Additionally, their analysis of the impact of social security spending showed no significant effect on growth.

According to some writers, evaluating the sample of rich countries might be beneficial since those countries generally have large public expenditures. Hence, this examination can show whether the increase in government expenditures has negative growth effects or not. In that

context, Fölster and Henrekson (2001) aimed to understand the relationship between government spending and growth by using a sample of rich countries between 1970 and 1995. Writers used the extreme bounds analysis method to test this relationship, and according to them, when more econometric problems are addressed in the research, the findings are more likely to be robust. Their findings suggested that a robust and negative relationship between government spending and economic growth exists. However, Agell et al. (2006) criticized Fölster and Henrekson's research by claiming that they ignored the issues of omitted variable bias, reverse causation, and non-random sample selection. Additionally, the criticized researchers responded that there is selection bias in all OECD data due to the similarity of country characteristics among OECD nations. They justified their research by stating that it is not feasible to eliminate simultaneity in growth equations because of the challenge of identifying a suitable instrument (Fölster & Henrekson, 2006).

Schaltegger and Torgler (2006) examined the relationship between public expenditure and economic growth by using the panel data of the 26 cantons within Switzerland between the 1981 and 2001 period. At the very beginning of the study, the authors acknowledge the possible negative relationship between public spending and growth, which stems from the large public sector that tends to exist in rich countries. The authors initially employed OLS, Random Effects, and fixed-effects models before conducting the Breusch-Pagan and Hausman tests. Conduction of the tests led to the determination that the fixed-effect method was the most suitable. As a result, their study found that in the case of wealthy states, a decrease in government current spending leads to a significantly positive growth rate. In contrast, they also found that an increase in government investment spending had no significant effect.

Stansel (2008) looked at the relationship between local government expenditure and economic growth using a dataset of US metropolitan areas for the period 1960 to 1990. He applied an OLS method and found that there was no significant relationship between local government expenditure and growth. However, he established that both investments by local government and investment in highways had a positive and significant correlation with growth. Similarly, Guerrero and Parker (2012) examined the causal link between government expenditure and GDP growth in the United States for the period from 1791 to 2009. The authors utilized time series

data and applied the Vector Error Correction Model (VECM). They concluded that there was no evidence to support a negative correlation between growth and government expenditure.

Holt et al. (2012) investigated the impact of expenditure by state governments on the growth of Gross Domestic Product (GDP). The researchers analyzed panel data from 50 US states over a period of 6 years and employed a fixed-effects method in their empirical model. The results indicated a significant positive relationship between state government expenditures and GDP growth. The researchers also examined the effect of political party management on growth but found no significant impact.

Checherita-Westphal and Rother (2012) investigated the impact of government debt on GDP growth by using the panel data of 12 Euro-area countries that covered the period between 1970 and 2008. The authors used a quadratic functional form to state the debt growth relationship more efficiently, and they employed panel fixed-effects and instrumental variable methods with two-stage least squares (2SLS) and GMM estimators to mitigate the simultaneity bias. It is important to note that the researchers instrumented their debt variable on the model with the time-lagged version of the debt or the average debt levels of other sample countries. Consequently, they found that there is a non-linear relationship between government debt and growth. Their findings suggested that above the range of 90 and 100 percent of debt to GDP ratio, any increase in debt ratios negatively affects the long-term growth rates.

The role of public debt on GDP growth is critical to understand, as it could affect productive government spending, which plays an important role in long-term growth (Teles & Mussolini, 2014). To capture the relationship between public debt and the effectiveness of fiscal policy, Teles and Mussolini (2014) proposed an endogenous growth model and used panel data from 74 countries for the period 1972 to 2004. The authors used a two-step Generalized Method of Moments (GMM) in their study and found that the effect of productive spending on growth depends on the debt-to-GDP ratio, and an increase in debt could lead to positive growth. However, they noted that a higher debt-to-GDP ratio reduces the effect of productive spending on growth. The authors also indicated that the relationship between productive spending and growth becomes negative when the debt-to-GDP ratio exceeds 22.5 percent.

Macek (2015) studied the effects of taxation on the economic growth in the OECD countries between 2000 and 2011 by using the fixed-effects method. The writer benefited from the neoclassical growth theory when employing its empirical model. According to the author's hypothesis, the increased income and corporate tax have distortionary effects on growth because these factors can negatively affect private investment in the economy. As a result of his study, he found that increased government spending is negatively associated with GDP growth, and this impact stems from the fact that increased spending has crowding-out effects, which basically leads to higher taxation.

In their research, Kutasi and Marton (2020) examined the impact of public spending on the GDP growth per capita. Their dataset included the 25 European countries between the periods of 1996 to 2017. Authors employed many different econometric methods, such as fixed-effects, OLS, and first-differences with GMM estimator, and created their empirical model based on Barro's growth model. Accordingly, the writers' findings suggest that the effects of government spending on GDP growth depend on the type of spending, which is aligned with the theory suggested by Barro (1990). In that sense, they found that lagged values of education and health spending are positively related to GDP growth.

Connolly and Li (2016) researched the impact of government consumption spending, especially public social spending and public investment, on the economic growth of the 34 OECD countries that covered the period of 1995 to 2011. By being aware of the possible existence of endogeneity in their growth equation, the authors directly employed a GMM estimation method in their panel data model. They also took the lagged values of all their independent variables and employed them in their GMM model. Consequently, the authors found a significant negative effect of public social spending on GDP growth. On the other hand, they also found no significant effect of government consumption spending and public investment on economic growth.

3. Data and Empirical Methodology

Data from the OECD dataset covering the period from 2002 to 2021 have been utilized. A total of 10 developed OECD nations, including Luxembourg, Ireland, Norway, Switzerland, the United States, Denmark, the Netherlands, Australia, Austria and Iceland, are included in the data. While the sample of rich countries may lead to some bias in the results, this research can present specific and valuable implications for wealthy countries. A sample of wealthy governments also brings homogeneity to the analysis, allowing for more meaningful comparisons.

To ensure the accuracy of our model, we thoroughly examined the correlations between the independent variables. Therefore, we created a correlation matrix for the independent variables in our model. When we look at the correlations between the independent variables from Table 1, there are a few conspicuous correlations that can be caught at first glance. Those are the correlations of the Educ and GES variables with the GDPpc variable as well as the correlation between PSS and GES. In that context, the most considerable correlation is 58 percent, and it exists between the GDP per capita and the adult tertiary education level. In comparison, the other correlations correspond to 42 percent and 38 percent, respectively. If we think about the reason for the 58 percent correlation between the education level and GDP per capita, there are many possible explanations for it. For instance, increased tertiary education levels might mean that the country has more human capital, and therefore, it can positively affect GDP per capita by boosting productivity. It also can mean that rising education levels can increase the awareness of health and political-related issues, which might cause a more stable society, and this stability can lead to a rising GDP per capita. And possibly this correlation might stem from the fact that when the GDP per person increases it is easier for people to complete their formal education and pursue further degrees. However, the correlations that are mentioned are not that problematic since, usually, correlations above 70 per cent are seen as signals for the multicollinearity issue. To ensure objectivity, we used a metric called the Variance Inflation Factor (VIF) to determine whether our model suffers from multicollinearity. In general, a VIF value above 5 indicates the presence of multicollinearity. The VIF values for our independent variables indicate that none of the values are close to 5 (Table 2). Thus, we can confidently conclude that our model is not affected by multicollinearity. Consequently, we can proceed to analyze the impact of government spending on growth.

Table 1: Correlation Matrix of the independent variables

	PSS	GES	GDPpc	Educ	U	POP	GCF
PSS	1	0.38	-0.003	-0.13	0.16	-0.18	-0.23
GES	0.38	1	-0.42	-0.10	-0.15	0.03	-0.18
GDPpc	-0.003	-0.42	1	0.58	-0.003	-0.06	-0.02
Educ	-0.13	-0.11	0.58	1	0.19	0.33	0.06
U	0.16	-0.15	-0.003	0.19	1	0.13	-0.28
POP	-0.18	0.03	-0.06	0.33	0.13	1	-0.16
GCF	-0.24	-0.18	-0.02	0.06	-0.28	-0.16	1

Table 2: Variance Inflation Factor Values

Variable Name	VIF Value
PSS	1.61
GDPpc	3.89
GCF	1.35
GES	2.23
Educ	3.06
U	1.66
POP	1.57

In that regard, our empirical model is revealed as follows:

$$\ln \text{GDPpc}_{it} - \ln \text{GDPpc}_{it-1} = \beta_0 + \beta_1 \ln \text{PSS}_{it} + \beta_2 \ln \text{GDPpc}_{it-1} + \beta_3 \ln \text{GCF}_{it} + \beta_4 \ln \text{GES}_{it} + \beta_5 \text{Educ}_{it} + \beta_6 \text{U}_{it} + \beta_7 \ln \text{POP}_{it} + \omega_i + \delta_t + \varepsilon_{it}$$

Figure- 3: Empirical Model

The model uses $\ln \text{GDPpc}_{it}$ to represent the natural logarithm of GDP per capita expressed by purchasing power parity (PPP) in country i and year t. The left-hand side of the equation represents the growth of GDP per capita, which is the dependent variable of our empirical model. The right-hand side of the equation includes $\ln \text{PSS}_{it}$, which represents the natural logarithm version of public social spending in country i and year t. Public social spending is the primary

independent variable, and its magnitude and significance are critical for the research. The model also includes $\ln GDP_{pcit-1}$, the natural logarithm version of initial GDP per capita, which is used due to the convergence process in growth theory. Additionally, our model includes the natural logarithm versions of gross capital formation (GCF) and government education spending (GES) as supportive variables, which can play a crucial role in the growth process. Similarly, the control variables included in the model are adult education level (Educ), unemployment rate (U), and population (POP).

According to the characteristics of our regression model, we chose to utilize the two-way fixed-effect method in our econometric model. Therefore our model includes country ω_i and time δ_t dummy variables. These dummy variables help our model to take account of the unobservable factors that might come from the country and time-specific conditions. However, to test the effect of public social spending on growth in the best possible way, the most suitable panel data method should be utilized. Therefore, throughout the research process, we utilize Hausman, Lagrange Multiplier (LM), Breusch-Pagan (BP), and Wooldridge tests to decide on the most effective model to get practical estimators.

As a part of the analysis process of our regression, we first will create five different models, which are going to represent the OLS, Fixed-Effects (FE), and Random Effects (RE) models in a table. The first two models, respectively, will be the OLS and Fixed-Effects models without the control variables, while the last three models will be the OLS, Random Effects, and Fixed-Effects with the control variables. In that sense, the complementary part of the control variables will be assessed and will be seen whether those variables improved our models or not. After the evaluation of all models through the interpretation, firstly, the Breusch-Pagan Lagrange Multiplier test (i.e. LM test), then the Hausman test, and lastly, the Wooldridge and Breusch-Pagan (BP) tests are going to be applied.

Overall, understanding the reasons behind these econometric tests that are going to be utilized during our analysis is critical to comprehend the characteristics of our model. So, in the way of reaching our final econometric model, firstly, the LM test is going to be used to decide which model amongst the Random-Effects and OLS is better in the context of our dataset. Because the LM test is a valuable tool that helps researchers to choose between the OLS and Random Effects

model, and its null hypothesis suggests that there are no individual (country in our case) specific unobserved effects in our panel data model, i.e., country-specific effects are not correlated with the independent variables. In other words, if we fail to reject the null hypothesis, that means our data has no panel effect; hence, it is better to use OLS. Secondly, in the Hausman test, the main goal is to decide between the Random Effects and Fixed-Effects models. Assuming that we found a panel effect through the application of the LM test, the Random Effects is chosen over the OLS. After that, we need to decide between the RE and FE. Accordingly, the Hausman test's null hypothesis suggests that within entity error terms are uncorrelated with the independent variables. Therefore, the null hypothesis in the Hausman test basically means "the preferred model is RE." And if we succeed in rejecting the null, then we must proceed with the FE model. Thirdly, in the Wooldridge test, the objective is to understand if the auto-correlation in the related model exists or not. In that regard, the Wooldridge test's null hypothesis suggests that the serial correlation does not exist. And the failure to reject the null means the related model does not suffer from auto-correlation. Lastly, the BP test will assist us in figuring out whether our last model was affected by heteroskedasticity. In the BP test, the null hypothesis tells that the model does not suffer from heteroskedasticity.

In this paper, the FE model will be used after the sequential steps are taken in terms of diagnostic econometric tests. The related steps and analysis will be presented in the results section in a detailed way. Additionally, In the case of the fixed-effects model, Schaltegger and Torgler (2006) mention that although the usage of FE might present many advantages, such as less risk of simultaneity bias, it is still a challenging task to make non-biased estimations from the yearly panel data. Fölster and Henrekson (2001) state that endogeneity bias lies among the problems in the short period of panel data observations. While this is the case, in most growth studies, the Instrumental Variable (IV) method is one of the popular tools to eliminate the unwanted results that stem from endogeneity and simultaneity. Many researchers, such as Kneller et al. (1999), Fölster and Henrekson (2001), Connolly and Li (2016), Kutasi and Marton (2020), and Schaltegger and Torgler (2006) utilized an IV method in their research. However, according to Agell et al. (2006), growth regressions, by their nature, are bound to suffer from endogeneity and simultaneity problems since the independent variables in the growth equation are reciprocal to each other in the cause-and-effect sense. Therefore, it is nearly impossible to find suitable

instruments that will be exogenous to these problematic relations. Also, it is shown that the usage of the IV method without having a solid instrument is more likely to create more problems rather than solve them (Agell et al., 1999). In that regard, by accepting the limitations that come from the econometric methods and the nature of growth equations, our econometric model will not make use of the IV method since finding good instruments and applying them could make the process more problematic.

4. Data Descriptions and Summary Statistics

Table 3: Data Sources and Data Descriptions

Variable Name	Description	Year	Source of the Data
GDPpc	GDP per capita expressed in current international dollars (PPP)	2002-2021	World Development Indicators database, World Bank
PSS	Public Social Spending as % of GDP	2002-2021	Social Spending indicator, OECD
GCF	Gross Capital Formation as % of GDP	2002-2021	World Bank national accounts data
GES	Government Expenditure on Education as % of Total Government Expenditure	2002-2021	UNESCO Institute for Statistics
Educ	Tertiary Education Levels of Adults (% of 25-64 year olds)	2002-2021	Adult Education Level indicator, OECD
U	Unemployment Rate (% of Total Labor Force)	2002-2021	International Labour Organization, Projections database (ILOSTAT)
POP	Population (in Millions)	2002-2021	Population indicator, OECD

As it is shown in Table 3, our model consists of 7 variables. And the period of our variables in the dataset is yearly. Although we benefited from the International Labour Organization

and Unesco Institute databases to collect our data, we mostly utilized the OECD and World Bank databases.

When we evaluate the summary statistics that are given in Table 4, we can see that the POP, GDPpc, and Educ are among the variables that have a relatively high volatility.

Table 4: Summary Statistics

Variable Name	Observation	Mean	Std. Dev.	Min.	Max.
GDPpc	200	54908	19463	29032	131631
PSS	200	20.40	4.66	12.87	31.11
GCF	200	23.41	4.65	13.91	54.77
GES	200	5.74	1.29	2.98	8.6
Educ	200	36.85	6.62	23.99	53.71
U	200	5.3	2.04	2.25	15.45
POP	200	38.34	91.68	0.28	332.03

5. Results

Using a panel data set of ten developed OECD countries for 20 years, Table 5 shows the results of five different models. Accordingly, the first two models represent the OLS and Fixed-Effects models without the control variables. When we evaluate these models, the effects of public social spending, initial GDP per capita, and government education spending on GDP growth are negative, and each one of these variables is significant at the 1% level. However, the absolute values of point estimations of public social spending and the initial GDP per capita are relatively higher in the FE model than it is in the OLS model, while the value of point estimation of

government education spending in the OLS model is slightly higher. When one considers the fact that the FE 1 model includes both time and country dummy variables while the OLS 1 model only includes the country dummy, the mentioned differences in coefficients are not surprising. Therefore, these large differences in the coefficients might stem from the effects of the business cycle since the time time-dummy variables are not included in the OLS 1 model. In addition to that, the FE 1 model has a bigger adjusted-R squared value when compared to the OLS 1 model, so it is also a signal that the usage of the fixed effects might be a more useful approach considering the characteristics of our dataset. In that regard, the coefficient of public social spending in the first FE model is -0.24, which means that a 10 percent increase in the share of public social spending in the GDP reduces the GDP growth by 2.4 percent on average. For the OLS 1 model case, the coefficient of public social spending corresponds to -0.099, which means that a 10 percent increase in the share of public social spending decreases the growth by 0.9 percent on average.

Table 5: Regression Estimation Results for Different Panel Models

Dependent Variable: GDP Growth per person					
	OLS 1	FE 1	OLS 2	RE	FE 2
Logarithm of PSS	-0.099*** (0.037)	-0.246*** (0.035)	-0.140*** (0.041)	0.011 (0.018)	-0.263*** (0.039)
Lagged logarithm of GDP per person	-0.045*** (0.016)	-0.376*** (0.048)	-0.134*** (0.032)	-0.063*** (0.021)	-0.361*** (0.049)
Logarithm of GCF	0.001 (0.024)	0.021 (0.021)	0.019 (0.028)	0.007 (0.020)	0.031 (0.023)
Logarithm of GES	-0.168*** (0.037)	-0.122*** (0.028)	-0.159*** (0.036)	-0.082*** (0.021)	-0.118*** (0.028)
Adult tertiary Education Level (% of 25-64 year-olds)	-	-	0.003** (0.001)	0.002** (0.001)	0.001 (0.001)
Unemployment Rate	-	-	0.001 (0.003)	-0.004** (0.002)	0.003 (0.002)
Logarithm of Population	-	-	0.094 (0.119)	-0.005** (0.002)	-0.096 (0.092)
Observations	190	190	190	190	190
R²	0.260	0.385	0.309	0.121	0.398
Adjusted R²	0.205	0.265	0.245	0.088	0.266
F-Statistic	4.756***	24.754***	4.837***	25.167***	14.660***

Notes: Standard errors in parentheses, and * p<0.1; ** p<0.05; *** p<0.01

Furthermore, when we examine the second models of the OLS and the FE, it can be seen that the addition of the adult education levels, unemployment rate, and the population as control variables into the models leads to some improvements in terms of adjusted-R squared values. In the second OLS model, it can be seen that the effect of adult education levels on growth is positive and significant at the 5% level. However, the coefficient of education is very close to zero; hence, its effect is not that meaningful. Additionally, the coefficients of public social spending and initial GDP per capita experienced a considerable rise, while the reverse was true

for government education spending. In the case of the FE model, the addition of the control variables to the model led to a slight reduction in the coefficients of initial GDP per capita and government education spending. In contrast, the coefficient of the public social spending experienced a rise.

On the other hand, the RE model in Table 5 shows a positive relation between public social spending and GDP per capita growth. However, the coefficient of public social spending in the RE model is 0.011, which is relatively low, and the estimation is not statistically significant. Moreover, when we investigate the coefficients of other variables in the RE model, it can be seen that the effects of initial GDP per capita, government education spending, unemployment rate, and population size are negative and statistically significant. However, the coefficients of the control variables are close to zero, and the coefficients of initial GDP per capita and government education spending are relatively lower compared to the OLS and FE models.

After analyzing the five models presented in Table 5, some conflicts were identified between the OLS, RE, and FE models in terms of their econometric results and implications. Therefore, to ensure an efficient analysis and meaningful economic outcomes, it may be necessary to consider and select the most suitable model carefully.

Table 6: Diagnostic tests between the models

Diagnostic Tests	P value	Null Hypothesis
Lagrange Multiplier Test	0.0216	No Panel Effect
Hausman Test	0.00000000000001808	Preferred Model is RE
Wooldridge Test	0.01653	No Serial Correlation
Breusch-Pagan Test	0.0002457	No Heteroskedasticity

Notes: if $p\text{-value} < 0.05$ the null hypothesis will be rejected

In Table 6, the p values and the null hypothesis of the diagnostic tests are given. In order to decide between the OLS and RE models, we will first utilize the LM test results from Table 6. According to the LM test results, we found the P value to be 0.02, which means that we reject the null hypothesis that suggests that our country variables are not correlated with our explanatory

variables. Hence, panel effects exist in our model, and random effects should perform better than the OLS. Secondly, after deciding that the RE model performs better than the OLS, we need to apply the Hausman test to choose between the RE and FE models. Since the result of the Hausman test is close to 0, we decided to continue with the FE model.

Although we have chosen to use FE in our research as our final model, and also knowing that it is one of the most preferred methods in the literature on growth research, some possible problems can arise from using it in the growth equations. First of all, it is known that fixed effects are beneficial since this method accounts for the unobserved country and time-specific factors. So, the advantage of taking the unobserved factors into account helps us to eliminate any possible omitted variable bias. However, the common problems in employing the fixed effect method might arise from the correlations between the country-specific conditions and the independent variables due to the characteristics of the growth equations, and it can also occur from independent variables themselves being endogenous (Connolly & Li, 2016). Along with these reasons, we need to check the characteristics of our FE model by utilizing other diagnostic tests.

As shown in Table 6, we utilized the Wooldridge and Breusch-Pagan tests to learn if our FE model suffers from auto-correlation and heteroskedasticity issues since those two problems might affect the statistical significance of our estimation results because they lead to biased standard errors and can negatively affect the efficiency of the coefficient estimates. As a result of the Wooldridge test, we found a P value of 0.01, which concludes that our model suffers from serial correlation. In addition, when we evaluate the P value of 0.0002 from the Breusch-Pagan test, it shows that our model is seriously exposed to heteroskedasticity. In light of these results, the regression results that are shown in Table 5 might be misleading, and those results can be considered insignificant since the findings of Wooldridge and BP tests proved that our FE model has auto-correlation and heteroskedasticity issues. To address these issues, the Arellano standard error estimator will be used. This estimator is considered one of the Heteroskedasticity and Autocorrelation Consistent (HAC) estimators and is often recommended when dealing with heteroskedasticity and autocorrelation in the fixed-effects model. The Arellano standard error estimator adjusts standard errors for heteroskedasticity and autocorrelation, resulting in statistical

inferences that are more robust and valid. Therefore, employing this HAC estimator in our model will improve the econometric efficiency of our research.

Table 7: Adjusted Standard Errors for Heteroskedasticity and Auto Correlation

Dependent Variable: GDP Growth per person				
Adjusted Fixed Effects Model				
	Estimate	Std. Error	t value	Pr(> t)
Logarithm of PSS	-0.263	0.08557485	-3.0787	0.002459 ***
Lagged logarithm of GDP per person	-0.361	0.06802074	-5.3113	0.00000037 ***
Logarithm of GCF	0.031	0.01453762	2.1550	0.0327 **
Logarithm of GES	-0.118	0.04625701	-2.5502	0.011737 **
Adult tertiary Education Level	0.001	0.00134062	0.4370	0.662690
Unemployment Rate	0.003	0.00301118	1.1494	0.252181
Logarithm of Population	-0.096	0.13934896	-0.6895	0.491513

Notes: * p<0.1; ** p<0.05; *** p<0.01

Accordingly, after adjusting our standard errors by utilizing the Arellano method, the regression results of our final FE model are given in Table 7. When the results of Table 7 are investigated, it can be seen that the standard errors of PSS, GDPpc, GES, Educ, U, and POP are enormously increased when we employ the HAC estimator in the model. This is one of the critical points to be aware of in the regression estimations because when econometric problems such as heteroskedasticity and autocorrelation are present in the models, they can create biased standard errors. As it is in our case, almost all the standard errors of independent variables in the former FE model are underestimated compared to the model with robust standard errors. As a result, this underestimation in the standard errors led to bigger absolute t values, which essentially increased the probability of rejecting the null hypothesis and affected the statistical significance of our estimations in the former FE model. Interestingly, the GCF variable experienced the opposite

since its standard errors were overestimated. On the other hand, when we focus on the statistical significance of our variables in the adjusted FE model compared to the former model shown in Table 5, there are some profound changes. For instance, the estimate of the GES variable became statistically significant at 5% while it was statistically significant at 1% before. This means that when we correct our model by using robust standard errors, the impact of the share of education spending in government expenditure on growth becomes less significant. As opposed to this, the estimate of the GCF variable became statistically significant at the 5% level, while it had no significance in the former FE model. In other words, the impact of the share of Gross Capital Formation in GDP on Growth becomes statistically significant when we use robust standard errors in our fixed effects model. Following the analysis of Table 7, the main findings from our research can be summarized more clearly.

According to our results, we found that in the case of developed OECD countries, a 10 percent increase in the share of public social spending in GDP reduces the growth of GDP per capita by 2.6 percent on average while everything else is constant. This finding accords with Barro's (1990) growth model, which suggests that government consumption expenditures are negatively associated with GDP per capita growth since government consumption expenditures are considered non-productive spending. Because rising levels of government involvement introduce some unintended consequences in the economy, in particular, when the size of the economy is thought of as a pie, when the government starts to increase its expenditure in an unproductive way, that means the government occupies some parts of the pie, and this can create disincentives for private investments. Not only that, but if the government starts to increase its spending, this cost of spending creates the condition that it has to be met through taxation or debt. Eventually, this increased consumption by the government can prevent the potential private sector spending by the “crowding-out effect.” Additionally, we found that the 10 percent increase in the share of Gross Capital Formation in GDP increases the GDP per capita growth by 0.3 on average. This positive effect of the Gross Capital Formation is also expected, and it aligns with the findings of Topçu et al. (2020). As it is known, gross capital formation is directly related to capital accumulation, and any rise in gross capital formation can have a boosting effect on production and investments. But when we compare the effects of gross capital formation on growth, it is weaker than public social spending. Moreover, when we look at the effect of government education spending on the GDP per capita growth, the relationship is negative and significant.

That is, the 10 percent increase in the share of government education spending on total government spending is associated with a 1.1 percent decrease in the GDP per capita growth. Considering this, the negative association between government education spending and GDP growth is economically reasonable because the developed countries already have higher education levels than the poorer countries. Also, the marginal effect of government education spending is usually expected to be negative since rich countries already have a higher amount of human capital than is needed for their economy, and the effects of education are tricky to observe in the short term. Lastly, when we consider the coefficient result of the initial GDP per capita, we found that a 10 percent increase in the initial GDP per capita is associated with a 3.6 percent reduction in the GDP growth per capita on average. Our result on the initial GDP per capita is consistent with the convergence hypothesis of the neoclassical growth theories, which suggests that countries with higher initial GDP per capita tend to experience slower growth rates than countries with lower initial GDP per capita. However, according to Barro (1991), the convergence hypothesis in growth is only valid when poorer countries have a sufficient amount of human capital stock that can help them break the low-income threshold. For our case, the implications of the convergence hypothesis are not significantly important for our article, as our sample consists only of developed rich countries.

Ultimately, our fixed-effects regression results present many important findings that can be helpful in comprehending the impacts of government spending on the GDP growth per capita. In that sense, the results mentioned above might give valuable insights into the potential implications of government tools that can play a role in the growth process. Nevertheless, while it's important to recognize that academic research in the economic literature has the potential to provide valuable insights into social and economic phenomena through the use of advanced technical tools, we also need to be aware of the potential downsides as well as acknowledge them. As a matter of fact, acknowledging the limitations of the study can be just as valuable as addressing the potential implications of successful research, as it can provide readers with more information, which can lead to well-equipped further research in the field of interest. In the case of our panel data model, we reached a p-value of 0.0000000000001808 before as a result of the Hausman test (Table 6). Moreover, one of the possible practical implications of the Hausman test is that it can give information about the possible endogeneity. Typically, when the results of the

p-tests are smaller than 0.05, it might be considered a signal for the endogeneity problem, and since, in our case, the p-value is even well below 0.01, which suggests that our model suffers greatly from the endogeneity problem. As we mentioned before, it is widespread for growth models to suffer from endogeneity due to the characteristics of the growth equation. Similarly, as stated by Schaltegger and Torgler (2006), studying how government spending affects economic growth can be challenging because of the two-way causality involved. It is important to take into account that government spending affects growth through supply-side mechanisms, while growth, in turn, affects government spending due to the income elasticity on the demand side.

6. Conclusion

In light of the above, this paper investigated the role of government spending on GDP per capita growth by exploiting the advantages of the panel dataset methods for the rich OECD countries. Our main findings suggest that any increase in the percentage share of public social spending in GDP could lead to a contraction in GDP growth significantly. This is consistent with Barro's (1990) predictions, which suggest that non-productive government expenditures like subsidies and the provision of goods and services introduce distorting effects on the economy. Also, the significant negative effect of public social spending on GDP growth is confirmed by the empirical studies of some researchers, such as Connolly and Li (2016) and Fournier and Johansson (2016). By the same token, our results indicate that any increase in the share of government expenditure in total government spending reduces the GDP growth significantly. Interestingly, considering the spending on education as a productive government expenditure, the indication of an adverse effect on growth by the increasing education expenditure might be controversial. Conversely, the fact that the potential positive impacts of education expenditure are not visible in the short term makes this finding reasonable. Indeed, the positive potential implications of education spending on growth can only be understood in the longer term, and these results can also vary according to the related sample of the dataset. This result is also concordant with the findings of Boldeanu and Tache (2015), who found that expenditure on education had negative effects on GDP growth for the sample of 30 European countries.

On the other hand, since the main focus of this paper is to understand the impact of public social spending on GDP per capita growth, the negative associations of initial GDP per capita and gross capital formation on growth are not substantially crucial for this research. Nonetheless, the

negative role of initial GDP per capita and the positive effect of gross capital formation on GDP growth are very aligned with the related growth theory.

Overall, one of the most important limitations of our research comes from the existence of endogeneity in our empirical model. In the growth models, it is highly likely that this endogeneity might stem from the nonrandom sample selection, omitted variables, and reverse causality (Agell et al., 2006). Even though using a fixed-effects estimation is one of the ways to deal with endogeneity problems, it is not a solution for reverse causality. Considering this limitation, the most important inference that can validly be taken from our research is that further researchers must acknowledge the probable existence of endogeneity in their study. They should primarily design their empirical model to first deal with those issues because this is the only way to get valid statistical inferences from the growth studies. Consequently, our research suggests that any increase in public social spending and education expenditures by developed nations causes lower growth rates in the short term. However, any practical policy implications that can be inferred from our research should be taken with a bit of a grain of salt due to the possible reverse causation between government spending and GDP per capita growth.

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